AI for Next-Generation Batteries — Results Report

Hybrid 1D-CNN + BiLSTM model for battery SOC / SOH / RUL estimation and risk classification, generated from the impedance-spectroscopy dataset (battery.csv).

Objectives

• Extract and pre-process battery aging data (impedance, current-ratio parameters).
• Develop a deep learning model capturing spatial (CNN) and temporal (BiLSTM) features.
• Estimate State of Health (SOH) and State of Charge (SOC) in real time.
• Predict Remaining Useful Life (RUL) and classify risk into 5 bands.
• Provide precautionary guidance per risk band for battery management.
• Compare the hybrid model against CNN-only, BiLSTM-only and classical ML baselines.

Dataset Summary

Samples	2160
Average SOH	69.50%
Average SOC	22.40%
Average RUL	168.7 cycles

Risk Category Distribution

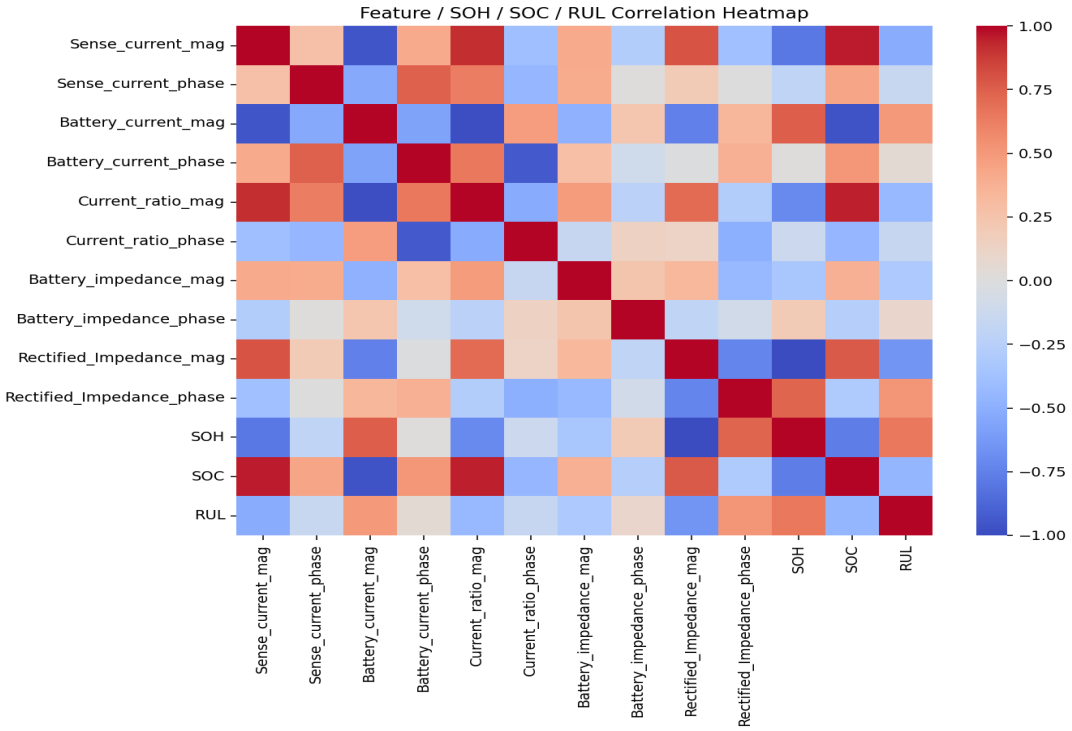
Risk Category	Samples
Excellent	848
Good	712
Low Risk	300
High Risk	253
Critical	47

Algorithm Performance Summary

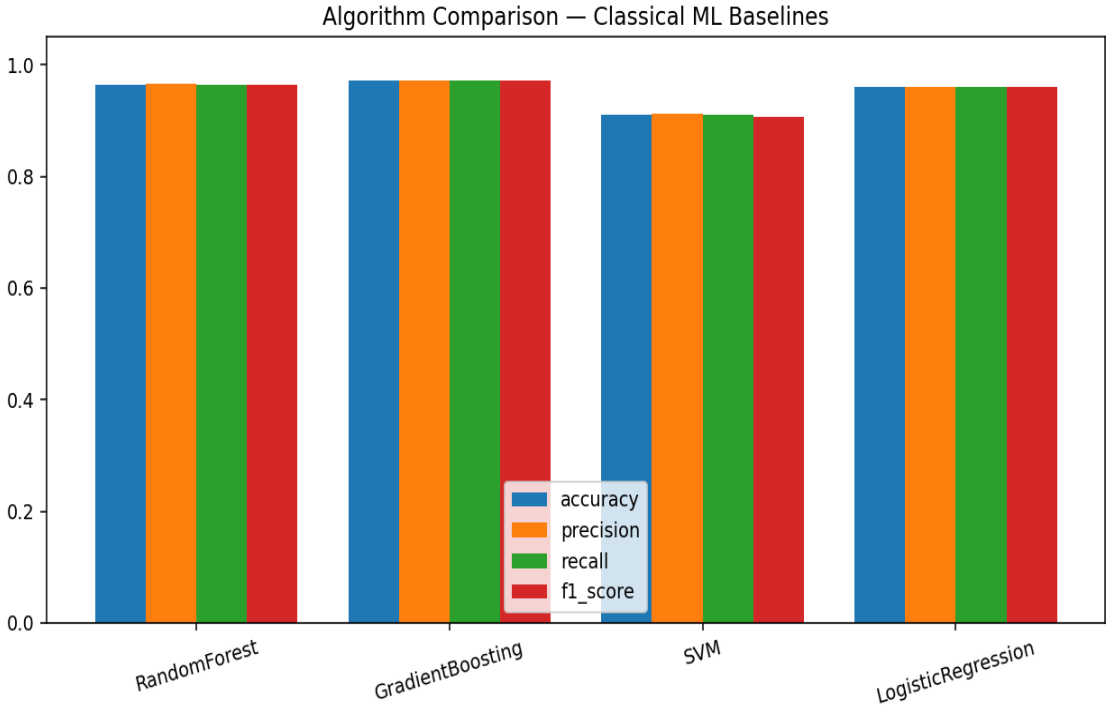
Algorithm	Accuracy	Precision	Recall	F1	Mean AUC
RandomForest	0.965	0.9655	0.965	0.9647	0.9988
GradientBoosting	0.972	0.9721	0.972	0.9719	0.9988
SVM	0.9112	0.9133	0.9112	0.9076	0.9937
LogisticRegression	0.9603	0.961	0.9603	0.96	0.9988

Visual Results

Correlation Heatmap

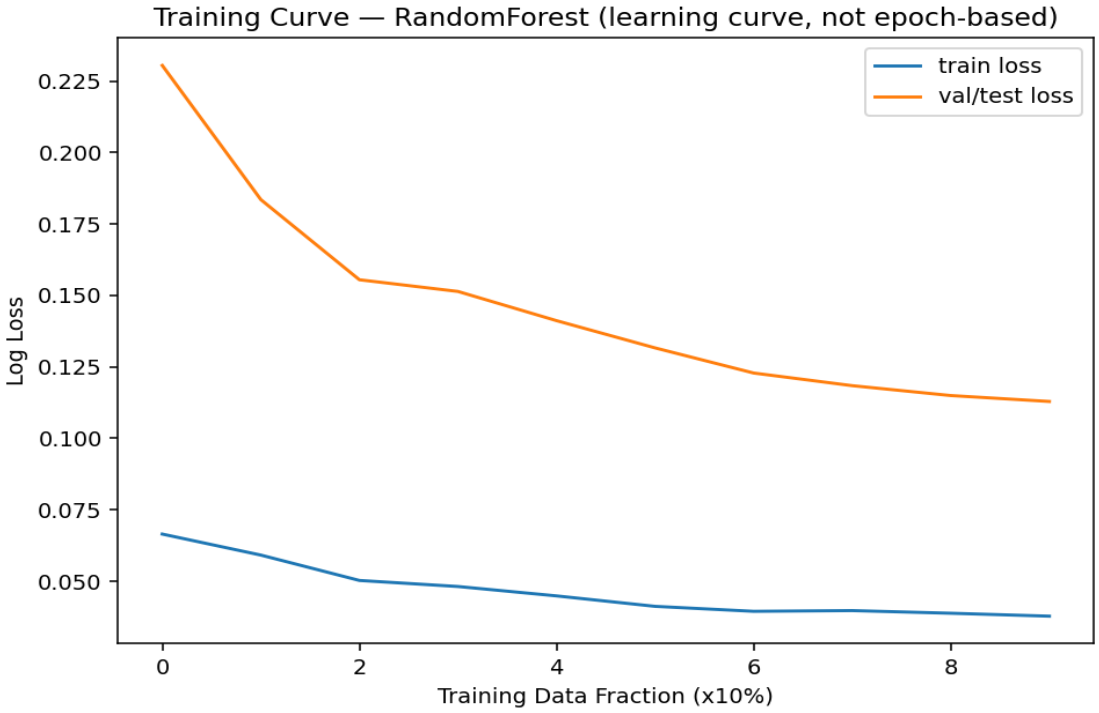


Algorithm Comparison (Classical Baselines)

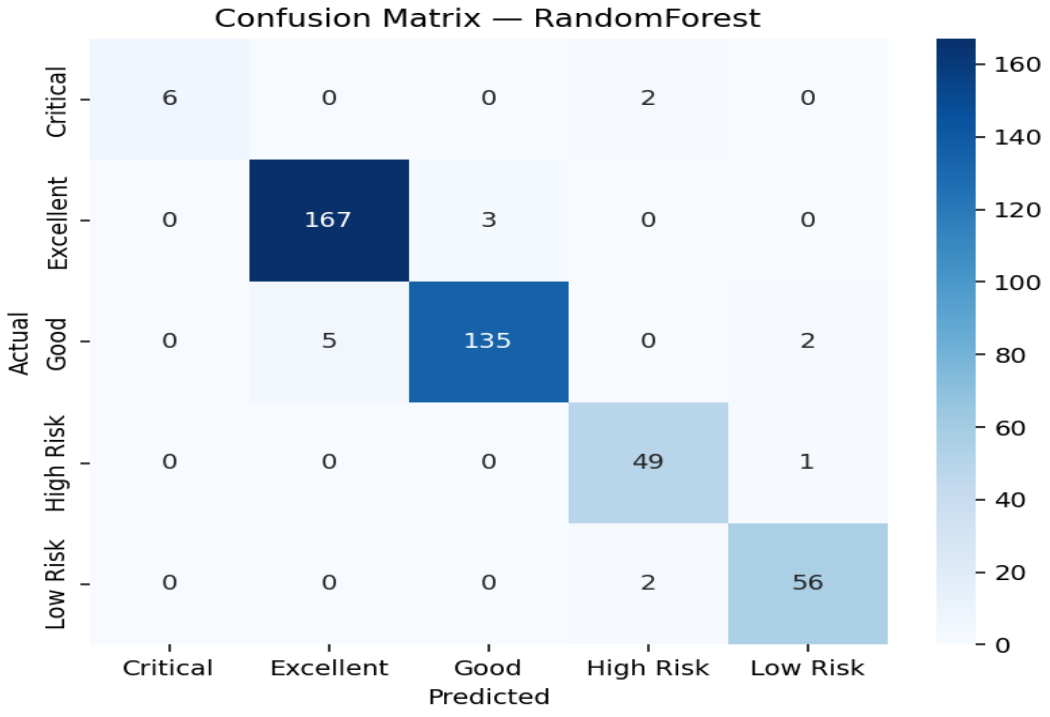


Algorithm: RandomForest

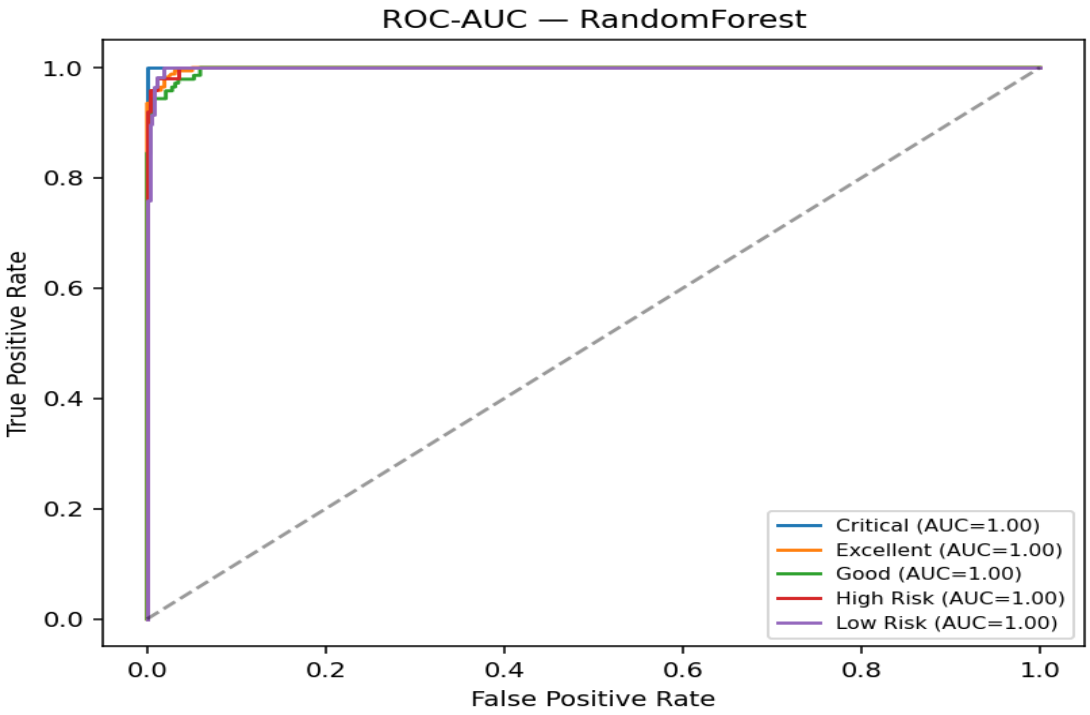
Training Curve — RandomForest



Confusion Matrix — RandomForest

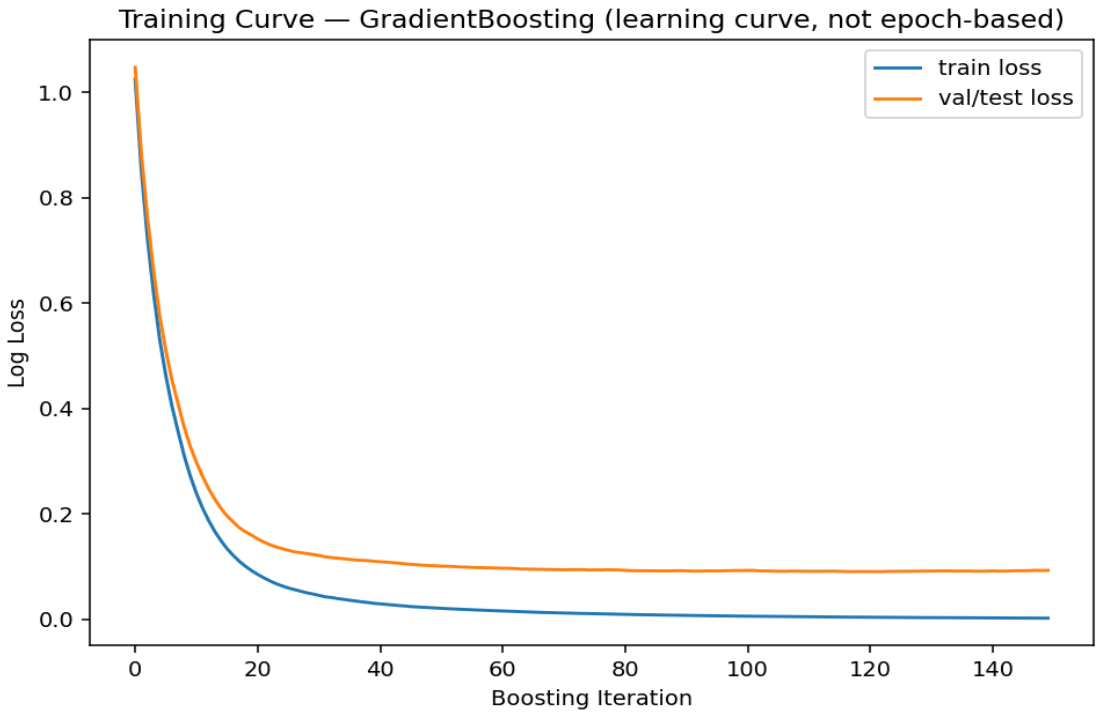


ROC-AUC — RandomForest

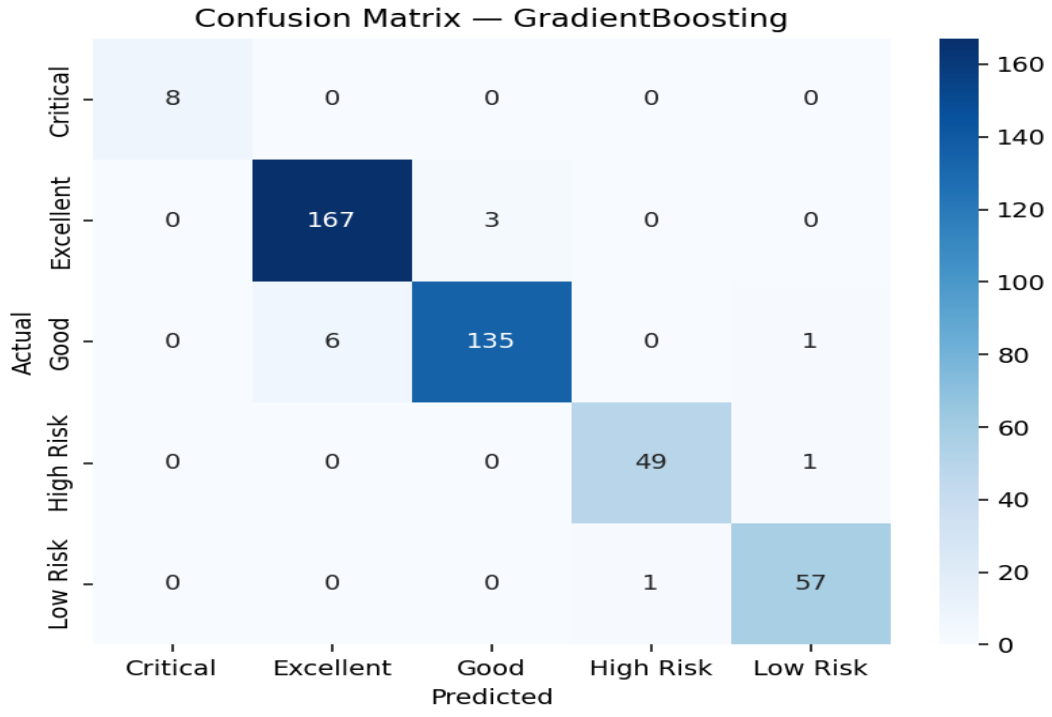

ROC-AUC — RandomForest
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
Critical (AUC=1.00)
Excellent (AUC=1.00)
Good (AUC=1.00)
High Risk (AUC=1.00)
Low Risk (AUC=1.00)

Algorithm: GradientBoosting

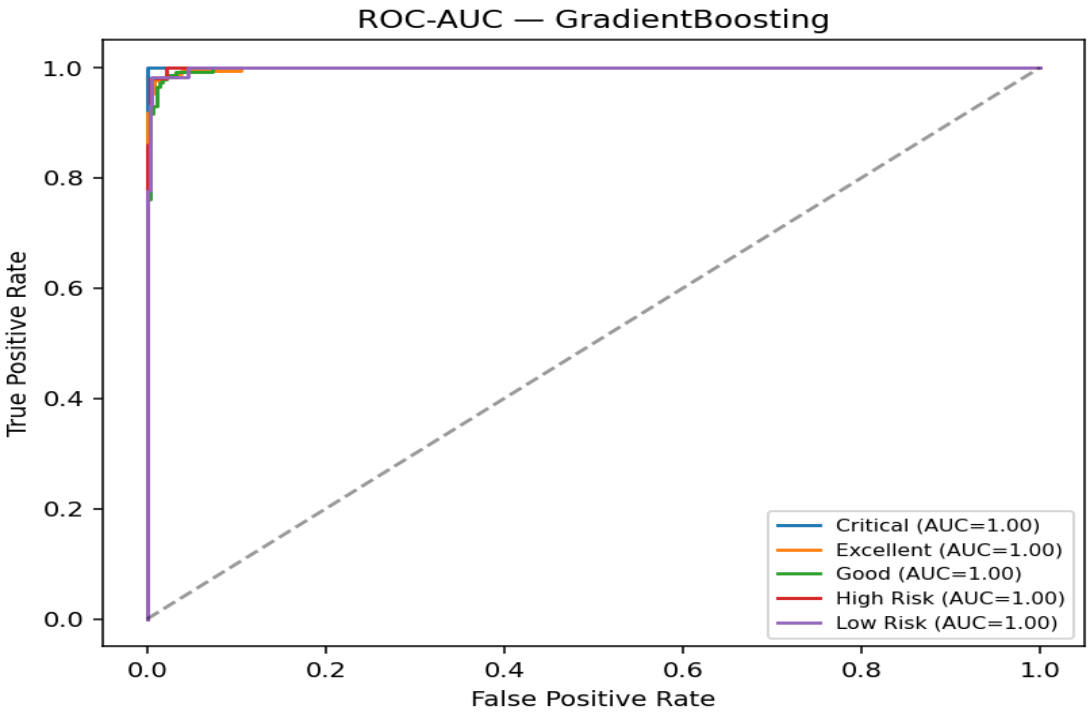
Training Curve — GradientBoosting



Confusion Matrix — GradientBoosting

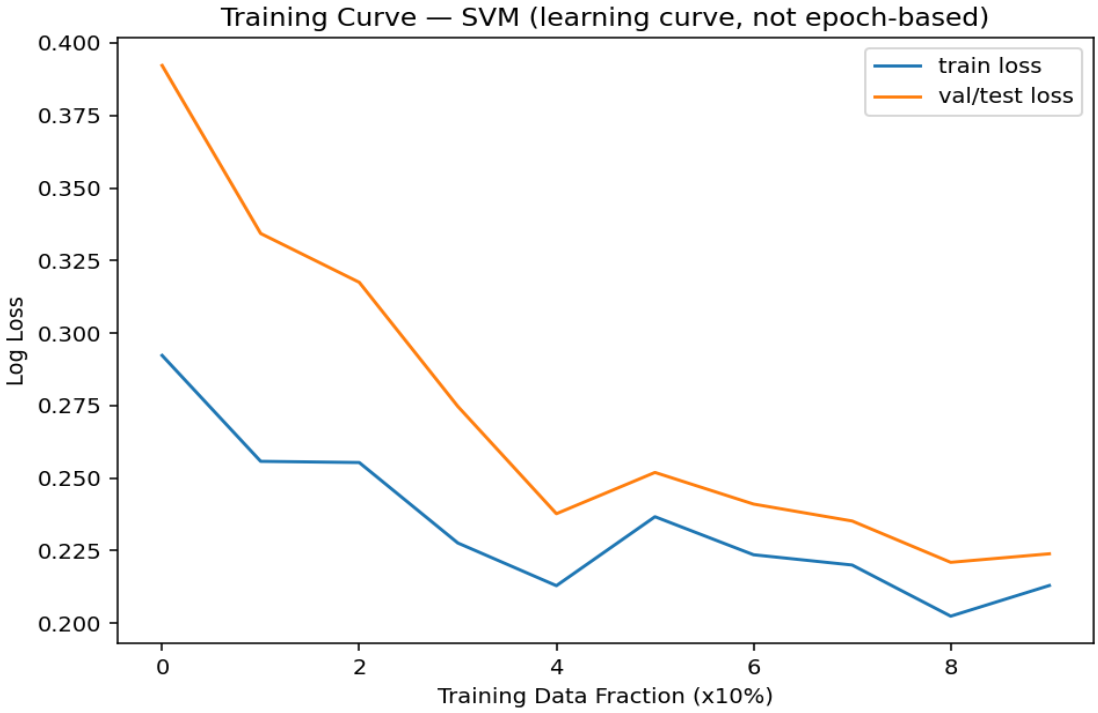


ROC-AUC — GradientBoosting

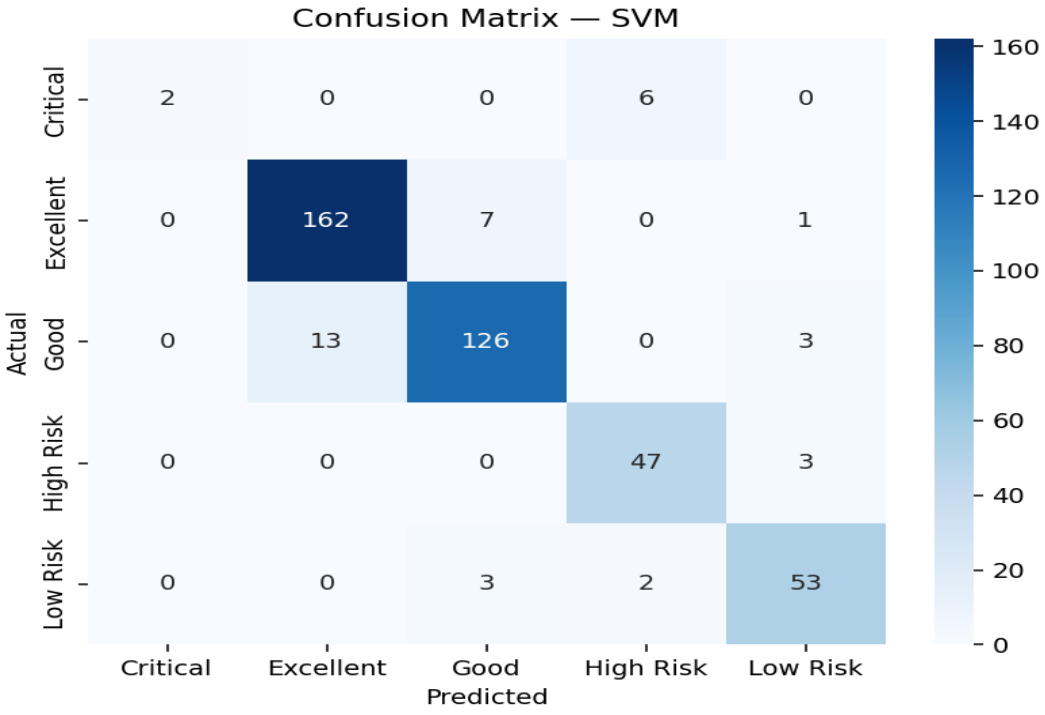

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Algorithm: SVM

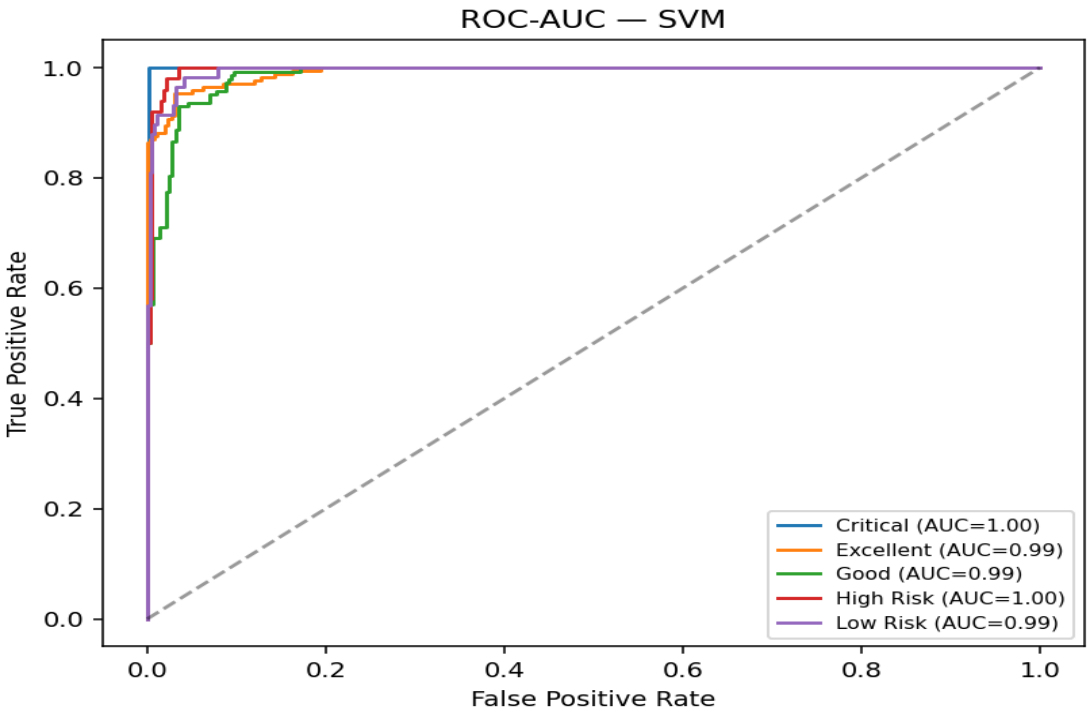
Training Curve — SVM



Confusion Matrix — SVM

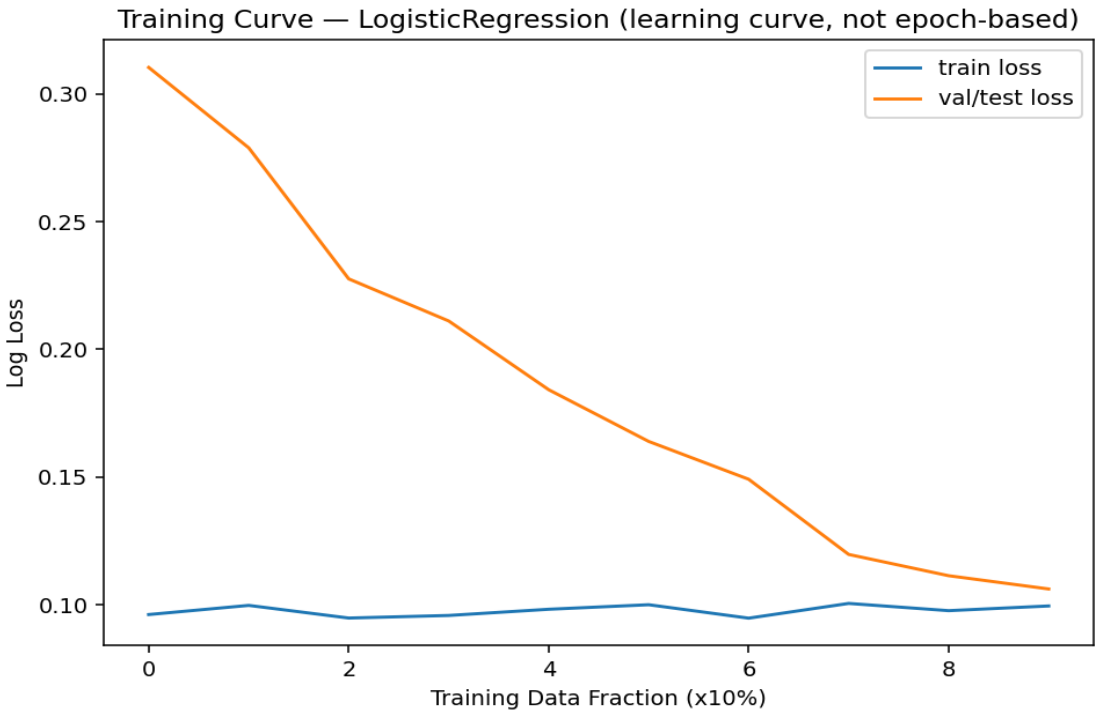


ROC-AUC — SVM

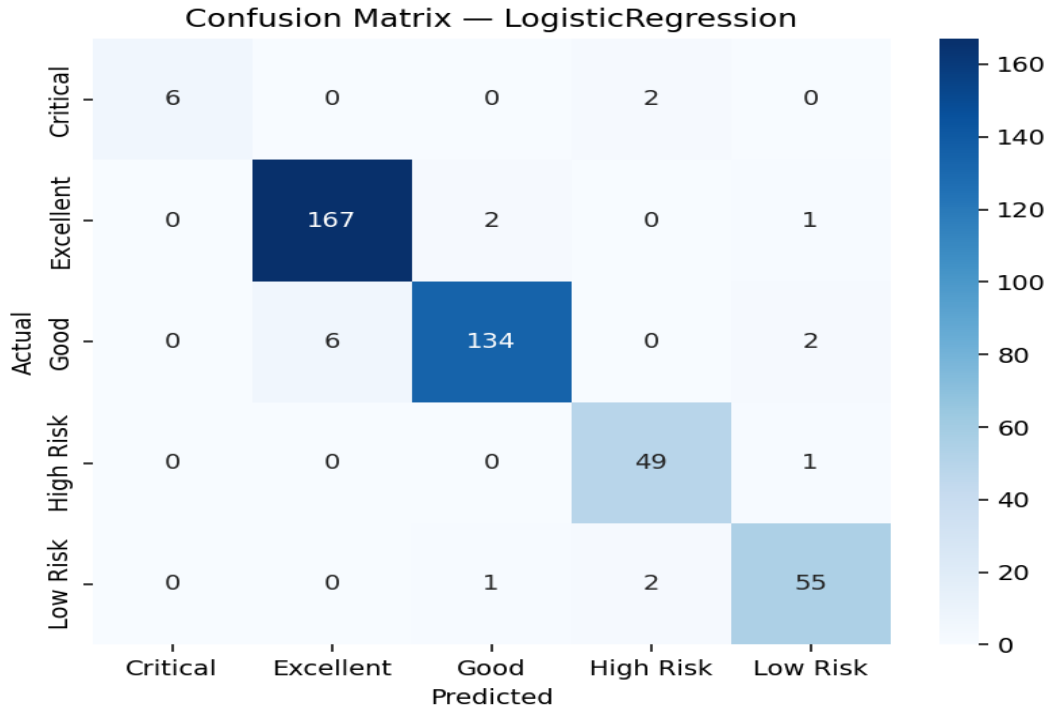

ROC-AUC — SVM
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
Critical (AUC=1.00)
Excellent (AUC=0.99)
Good (AUC=0.99)
High Risk (AUC=1.00)
Low Risk (AUC=0.99)

Algorithm: LogisticRegression

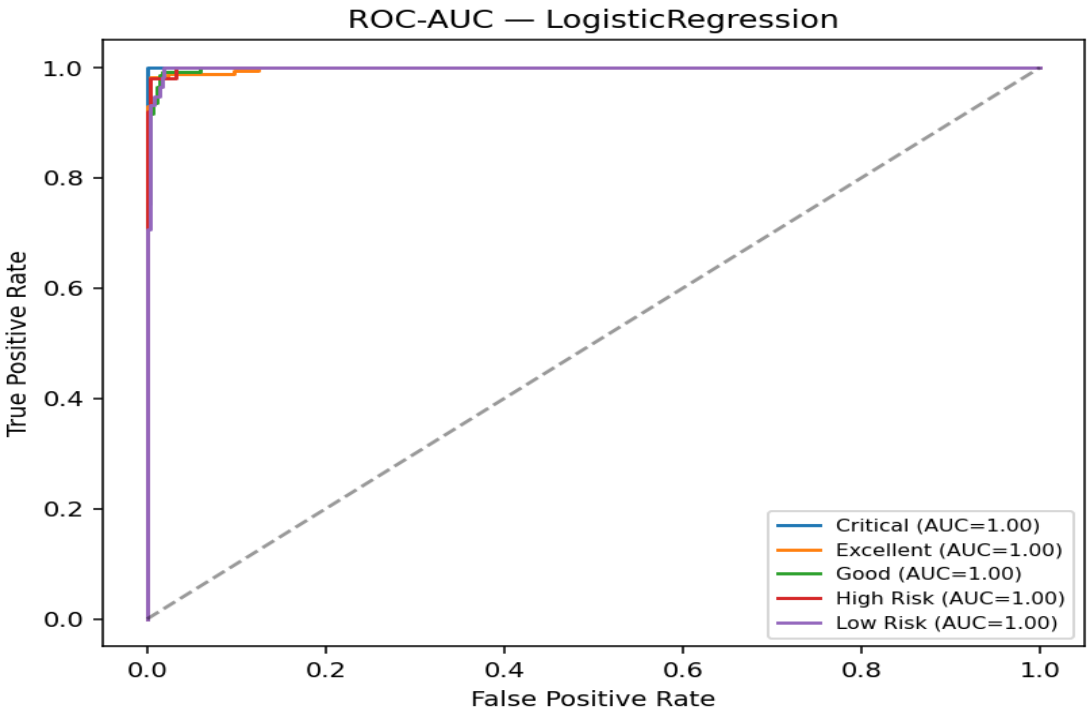
Training Curve — LogisticRegression



Confusion Matrix — LogisticRegression



ROC-AUC — LogisticRegression


ROC-AUC — LogisticRegression
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
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1.0
Critical (AUC=1.00)
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